Hermiticity and Mass Generation in the Chiral Dirac Equation

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Abstract

We examine the physical consistency and phenomenological implications of the Chiral Dirac Equation (CDE), a generalization of the Dirac field arising from the irreducible representations of the Poincaré group. By explicitly constructing the Hamiltonian operator, we demonstrate that the theory retains strict Hermiticity, ensuring unitary time evolution despite the inclusion of a chiral rotation parameter $\boldsymbol{\alpha}$ in the mass term. Addressing the neutrino mass generation problem, we implement a Chiral Seesaw Mechanism where the interplay between the chiral Dirac mass $\boldsymbol{me^{-i\alpha}}$ and a heavy Majorana scale $\boldsymbol{M_R}$ yields natural mass suppression. Crucially, this mechanism identifies the chiral angle as a geometric source of leptonic CP violation via the phase $\boldsymbol{e^{-2i\alpha}}$. Furthermore, we extend this formalism to the bosonic sector using the Bargmann-Wigner method. Our derivation suggests a universal mass scaling law $\boldsymbol{M_{\text{eff}} \approx m\cos^{2s}(\alpha)}$ applicable to Proca ($\boldsymbol{s = 1}$) and Rarita-Schwinger ($\boldsymbol{s = 3/2}$) fields. We conclude that the chiral degrees of freedom are intrinsic to the algebraic structure of relativistic wave equations, providing a unified description of mass modulation and symmetry breaking across arbitrary spin sectors.

Keywords: Chiral Dirac Equation, Chiral Seesaw Mechanism, Bargmann-Wigner Formalism, Leptonic CP Violation

1 Introduction

The foundational role of group theory in Quantum Field Theory (QFT) was established by Wigner [1], who showed that physical states are defined as unitary irreducible representations of the inhomogeneous Lorentz group (Poincaré group). This classification, based on the "little group" method, reveals that fundamental wave equations emerge as specific realizations of the underlying symmetry structure: scalar fields (spin 0) are governed by the Klein-Gordon equation, while vector fields (spin 1) correspond to the Maxwell and Proca equations [2]. Within this framework, the Dirac equation occupies a role for describing fermionic fields; while the modern group-theoretical approach derives it directly from the transformation properties of spinors under the Lorentz group, its original formulation was motivated by the necessity to resolve "duplexity" phenomena in atomic spectra, leading Dirac to construct a wave equation linear in time and momentum that satisfied the requirements of both relativity and quantum mechanics [3].

In recent developments, Watson and Musielak [4–6] revisited the group-theoretical foundations of the field description. Instead of relying on the standard factorization of the Klein-Gordon equation, they derived the field equations directly from the irreducible representations of the Poincaré group extended by parity. This approach revealed an intrinsic freedom in the choice of the chiral basis, parametrized by a new physical degree of freedom identified as the chiral angle. By identifying this degree of freedom, they obtained a generalized form of the Dirac equation, hereafter referred to as the Chiral Dirac Equation (CDE), which incorporates a pseudoscalar mass term alongside the standard scalar mass. The authors suggested that this additional chiral degree of freedom has physical implications; specifically, the interplay between the scalar and pseudoscalar mass components offers a mechanism to explain the suppression of neutrino masses, proposing the new massive particle as a candidate for Dark Matter.

The first critical issue concerns the consistency of the quantum mechanical description derived from the CDE. Standard quantum mechanics postulates that physical observables must be represented by self-adjoint operators to guarantee real energy eigenvalues and unitary time evolution (probability conservation). While the Dirac equation is consistent with these principles, the introduction of a chiral angle and a pseudoscalar mass term modifies the structure of the evolution operator. It is not immediately evident that the Hamiltonian associated with this generalized equation retains hermiticity under the new chiral basis transformations. Therefore, we aim to explicitly derive the Hamiltonian from the CDE and test its hermiticity to ensure the theory's physical viability.

Secondly, a significant algebraic inconsistency arises when applying the framework to neutrino physics under strict chirality constraints [4]. Preliminary analyses indicate that for a pure left-handed field, defined by the chiral projection $\nu_L = \frac{1}{2}\left(1-\gamma^5\right)\nu$, the scalar bilinear $\overline{\nu}\nu$ vanishes identically. Consequently, the Yukawa Lagrangian $\mathcal{L}_Y$ derived in the original work becomes trivial ($\mathcal{L}_Y = 0$), thereby nullifying the proposed mechanism for generating massive neutrinos in the Dirac context. To address this, we propose to explore the Majorana mass term within the context of a more general see-saw mechanism, functioning as a chiral see-saw.

Finally, while the original derivation focused on bispinors, effectively restricting the analysis to spin-1/2 particles [4], Watson et al. have also explored vector bosons [6]. However, the method employed—based on the irreducible representations of the Poincaré group—is fundamentally geometric and should be applicable to other spin sectors. To this end, we propose a review of the feasibility of extending this chiral formalism to higher-spin fields. Specifically, we examine whether similar chiral degrees of freedom can be consistently introduced into the Proca equation for massive vector bosons (spin-1) or the Rarita-Schwinger fields (spin-3/2).

2 Theoretical Framework

We adopt the formalism of relativistic quantum field theory on a flat, non-interacting Minkowski spacetime. The background geometry is defined by the orthonormal basis $\{\gamma^\mu\}$ satisfying the Clifford algebra $\mathcal{C}\ell_{1,3}(\mathbb{C})$, defined by the fundamental anticommutation relation

$$\{\gamma^\mu, \gamma^\nu\} \equiv \gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu = 2\eta^{\mu\nu}, \tag{1}$$

where $\eta^{\mu\nu} = \mathrm{diag}(1, -1, -1, -1)$ is the Minkowski metric. Consequently, $(\gamma^0)^2 = 1$ and $(\gamma^k)^2 = -1$. Greek indices $\mu = 0, \ldots, 3$ denote spacetime coordinates, while Latin indices $k = 1, \ldots, 3$ denote spatial components.

The chiral structure of the theory is determined by the pseudoscalar element γ^5, defined as

$$\gamma^5 \equiv i\gamma^0\gamma^1\gamma^2\gamma^3. \tag{2}$$

Since the spacetime dimension is even, γ^5 anticommutes with all vector generators γ^μ and squares to the identity, $(\gamma^5)^2 = 1$. This algebraic property permits the construction of the orthogonal chiral projectors

$$P_L = \frac{1}{2}(1 - \gamma^5), \quad P_R = \frac{1}{2}(1 + \gamma^5), \tag{3}$$

which satisfy the idempotency relations $P_{L,R}^2 = P_{L,R}$ and the orthogonality condition $P_L P_R = 0$. A spinor field ψ therefore decomposes uniquely into left- and right-handed chiral components, $\psi = P_L\psi + P_R\psi \equiv \psi_L + \psi_R$.

2.1 The Chiral Dirac Ansatz

Standard formulations of the Dirac equation assume a scalar mass term that couples chiral sectors trivially. Following Watson and Musielak [4], we consider a generalized dynamical equation invariant under the irreducible representations of the Poincaré group extended by parity. This yields the Chiral Dirac Equation (CDE) for a spinor field ψ

$$(i\gamma^\mu\partial_\mu - me^{-i\alpha\gamma^5})\psi = 0. \tag{4}$$

Here, m represents the rest mass, and α is a chiral angle parametrizing the rotation between the scalar and pseudoscalar mass sectors. The corresponding Lorentz-invariant effective Lagrangian density is given by

$$\mathcal{L}_{\text{eff}} = \bar{\psi}(i\gamma^\mu\partial_\mu - me^{-i\alpha\gamma^5})\psi, \tag{5}$$

where $\bar{\psi} \equiv \psi^\dagger\gamma^0$ is the Dirac adjoint. This Lagrangian serves as the generating functional for the dynamics and conservation laws investigated in the subsequent sections.

3 Hamiltonian Consistency and Hermiticity

To ensure the physical viability of the Chiral Dirac Equation (CDE), we must verify that it generates a unitary time evolution. This requirement demands that the Hamiltonian operator governing the dynamics be self-adjoint on the Hilbert space of states. We proceed by explicitly deriving the Hamiltonian from the effective Lagrangian density and analyzing its properties under Hermitian conjugation.

3.1 Canonical Derivation

The canonical momentum conjugate to the field ψ is obtained from the Lagrangian density $\mathcal{L}_{\text{eff}}$ (Eq. 5) via differentiation with respect to the time derivative of the field, $\dot{\psi} \equiv \partial_0 \psi$

$$\pi = \frac{\partial \mathcal{L}_{\text{eff}}}{\partial \dot{\psi}} = i\psi^\dagger. \tag{6}$$

The Hamiltonian density $\mathcal{H}$ is constructed via the Legendre transform

$$\mathcal{H} = \pi\dot{\psi} - \mathcal{L}_{\text{eff}}. \tag{7}$$

Substituting the explicit form of $\mathcal{L}_{\text{eff}}$ and isolating the temporal component of the derivative term, we find

$$\begin{aligned}
\mathcal{H} &= i\psi^\dagger\dot{\psi} - \psi^\dagger\gamma^0(i\gamma^0\dot{\psi} + i\gamma^k\partial_k\psi - me^{-i\alpha\gamma^5}\psi) \\
&= i\psi^\dagger\dot{\psi} - i\psi^\dagger\dot{\psi} - i\psi^\dagger\gamma^0\gamma^k\partial_k\psi + \psi^\dagger\gamma^0 me^{-i\alpha\gamma^5}\psi \\
&= \psi^\dagger\left(-i\gamma^{0k}\partial_k + \gamma^0 me^{-i\alpha\gamma^5}\right)\psi,
\end{aligned} \tag{8}$$

where we have introduced the Hermitian spatial generators $\gamma^{0k} \equiv \gamma^0\gamma^k$. The quantum mechanical Hamiltonian operator H, defined by the Schrödinger-like evolution equation $i\partial_0\psi = H\psi$, is therefore identified as

$$H = \gamma^{0k}p_k + \gamma^0 me^{-i\alpha\gamma^5}, \tag{9}$$

where $p_k = -i\partial_k$ is the spatial momentum operator.

3.2 Proof of Hermiticity

For the theory to be consistent, H must be Hermitian ($H = H^\dagger$). We examine the adjoint of the derived operator

$$H^\dagger = \left(\gamma^{0k}p_k\right)^\dagger + \left(\gamma^0 me^{-i\alpha\gamma^5}\right)^\dagger. \tag{10}$$

The first term is the standard kinetic energy operator. Since the momentum p_k is Hermitian and the matrix γ^{0k} is Hermitian, as $(\gamma^0\gamma^k)^\dagger = \gamma^{k\dagger}\gamma^{0\dagger} = (-\gamma^k)(\gamma^0) = \gamma^0\gamma^k$, their product is Hermitian.

The second term, containing the chiral mass correction, requires careful algebraic treatment. Expanding the adjoint yields

$$\left(\gamma^0 m e^{-i\alpha\gamma^5}\right)^\dagger = m\left(e^{-i\alpha\gamma^5}\right)^\dagger \left(\gamma^0\right)^\dagger. \tag{11}$$

Utilizing the hermiticity of γ^5 and γ^0, and the identity $(e^A)^\dagger = e^{A^\dagger}$, we obtain

$$m e^{i\alpha\gamma^5}\gamma^0. \tag{12}$$

Critically, γ^0 and γ^5 anticommute ($\{\gamma^0, \gamma^5\} = 0$). This anticommutation implies the commutation relation for the exponential function

$$e^{i\theta\gamma^5}\gamma^0 = \left(\cos\theta + i\gamma^5\sin\theta\right)\gamma^0 = \gamma^0\left(\cos\theta - i\gamma^5\sin\theta\right) = \gamma^0 e^{-i\theta\gamma^5}. \tag{13}$$

Applying this commutation to the mass term, we recover the original operator

$$m e^{i\alpha\gamma^5}\gamma^0 = \gamma^0 m e^{-i\alpha\gamma^5}. \tag{14}$$

Thus,

$$H^\dagger = \gamma^{0k} p_k + \gamma^0 m e^{-i\alpha\gamma^5} = H. \tag{15}$$

We conclude that the Hamiltonian of the Chiral Dirac Equation is strictly Hermitian. The introduction of the chiral angle α preserves the self-adjoint nature of the energy operator, ensuring real energy eigenvalues and unitary time evolution without requiring a modification of the inner product or metric.

4 Chiral Seesaw Mechanism

As an alternative approach to explain the smallness of neutrino masses, we revisit the mass generation mechanism proposed by Watson and Musielak within the framework of the Chiral Dirac Equation (CDE) [4, 5]. In this formalism, the fermion mass arises from Yukawa couplings to both a scalar field ϕ_1 (Standard Model-like) and a pseudoscalar field ϕ_2. In Watson's proposal [6], restricting the theory to pure left-handed fermionic fields causes the mass terms to vanish identically. Considering the original Yukawa Lagrangian postulated by Watson

$$\mathcal{L}_Y = -\frac{\lambda_1 v_1}{\sqrt{2}}\,\overline{\nu}\nu - \frac{\lambda_2 v_2}{\sqrt{2}}\,\overline{\nu}\gamma^5\nu. \tag{16}$$

Both terms vanish if we impose the left-handed chirality constraint on the neutrino field ($P_L\nu = \nu$, $P_R\nu = 0$). The scalar product $\overline{\nu}\nu$ vanishes because the Dirac adjoint of a left-handed field behaves as a right-handed projector acting to the left

$$\begin{aligned}
\overline{\nu} &= \overline{P_L \nu} \\
&= \nu^\dagger P_L \gamma^0 \\
&= \nu^\dagger \gamma^0 P_R \\
&= \overline{\nu}\, P_R.
\end{aligned} \tag{17}$$

Multiplying by ν from the right, and recalling that $\nu = P_L \nu$, we obtain

$$\overline{\nu}\,\nu = (\overline{\nu}\, P_R)\,(P_L \nu) = \overline{\nu}\,\underbrace{(P_R P_L)}_{0}\,\nu = 0. \tag{18}$$

Similarly, for the pseudoscalar term $\overline{\nu}\gamma^5\nu$, we utilize the fact that γ^5 commutes with the chiral projectors. Following the previous result

$$\begin{aligned}
\overline{\nu}\gamma^5\nu &= (\overline{\nu} P_R)\,\gamma^5\,(P_L \nu) \\
&= \overline{\nu}\gamma^5\,(P_R P_L)\,\nu \\
&= 0.
\end{aligned} \tag{19}$$

Thus, without the inclusion of a right-handed component, no Dirac mass term can be generated in this framework. However, if we assume the existence of a right-handed neutrino component ν_R (such that $\nu = \nu_L + \nu_R$), the Dirac mass term does not vanish. Instead, it acquires a complex phase determined by the chiral angle α, which is intrinsic to the CDE. To naturally suppress the neutrino mass without relying on fine-tuning, we invoke the Majorana mass term, leading to a Chiral Seesaw Mechanism [7].

4.1 The Chiral Dirac Mass Term

Following Watson and Musielak [4], the general Yukawa Lagrangian expanded about the vacuum expectation values (VEVs) v_1 and v_2 corresponds to Equation (16). Defining the scalar mass parameter $M = \lambda_1 v_1/\sqrt{2}$ and the pseudoscalar mass parameter $\tilde{M} = \lambda_2 v_2/\sqrt{2}$, and decomposing the fields into their chiral components, the Dirac mass term becomes

$$\mathcal{L}_{\text{Dirac}} = \overline{\nu}_L \left(M + \tilde{M}\right) \nu_R + \overline{\nu}_R \left(M - \tilde{M}\right) \nu_L. \tag{20}$$

In the CDE formalism, these parameters are related to the propagation mass m and the chiral angle α by $M = m\cos\alpha$ and $\tilde{M} = -im\sin\alpha$. Consequently, the effective Dirac mass coupling M_D connecting $\overline{\nu}_L$ and ν_R is identified as

$$M_D \equiv M + \tilde{M} = m\,(\cos\alpha - i\sin\alpha) = me^{-i\alpha}. \tag{21}$$

The result in (21) is crucial, the Dirac mass in this model carries a physical phase derived from the underlying chiral symmetry breaking.

4.2 The Full Majorana-Dirac Lagrangian

To implement the seesaw mechanism, we introduce Majorana mass terms. Unlike Dirac terms which couple left- and right-handed fields ($\overline{\nu}_L \nu_R$), Majorana mass terms violate lepton number by two units. Specifically, we utilize the charge-conjugated right-handed field $\nu_R^c = C\overline{\nu}_R^T$. The most general Lorentz-invariant mass Lagrangian including both the Chiral Dirac term derived above and the Majorana terms is

$$\mathcal{L}_{\text{Mass}} = \overline{\nu}_L M_D \nu_R + \frac{1}{2}\overline{\nu_L^c} M_L \nu_L + \frac{1}{2}\overline{\nu_R^c} M_R \nu_R + \text{h.c.} \tag{22}$$

Here, M_L represents the Majorana mass for the active left-handed neutrinos (assumed to be negligible or zero in the minimal Type-I seesaw), and M_R is the heavy Majorana mass scale for the sterile neutrinos.

4.3 The Chiral Mass Matrix

To find the physical mass eigenstates, we rewrite (22) using the basis of left-handed fields. We define the interaction basis spinor $\Psi = (\nu_L, \nu_R^c)^T$. Utilizing the Majorana identity, the Lagrangian allows the explicit matrix expansion

$$\mathcal{L}_{\text{Mass}} = \frac{1}{2}\overline{\Psi^c} \mathcal{M} \Psi + \text{h.c.}, \tag{23}$$

where the mass matrix $\mathcal{M}$ incorporates the chiral modifications from the CDE defined in (21)

$$\mathcal{M} = \begin{pmatrix} M_L & M_D \\ M_D^T & M_R \end{pmatrix} = \begin{pmatrix} M_L & me^{-i\alpha} \\ me^{-i\alpha} & M_R \end{pmatrix}. \tag{24}$$

4.4 The Neutrino Chiral Mass

To account for the smallness of neutrino masses, we adopt the following identifications in the matrix (24)

1. M_R**:** Heavy sterile scale ($M_R \gg m$).
2. M_L**:** Assumed $M_L \approx 0$.
3. M_D**:** Given by $me^{-i\alpha}$.

Based on these assumptions, the mass matrix is given by

$$\mathcal{M} \approx \begin{pmatrix} 0 & me^{-i\alpha} \\ me^{-i\alpha} & M_R \end{pmatrix}. \tag{25}$$

Finding the eigenvalues of (25) and applying the seesaw approximation ($m \ll M_R$), we obtain two mass terms

1. **Heavy Mass (m'_1):** Sterile Dominant State.

$$m'_1 \approx M_R + \frac{m^2 e^{-2i\alpha}}{M_R} \approx M_R \tag{26}$$

2. **Light Mass (m'_2):** Active Dominant State.

$$m'_2 \approx -\frac{\left(me^{-i\alpha}\right)^2}{M_R} = -\frac{m^2}{M_R}e^{-2i\alpha} \tag{27}$$

4.5 Mass Suppression and CP Violation

The application of the Chiral Seesaw Mechanism yields the following expression for the active neutrino mass based on (27)

$$m'_\nu \approx \frac{m^2}{M_R}e^{-2i\alpha}. \tag{28}$$

This equation implies two significant physical results.

4.5.1 Mass Suppression

The magnitude of the active neutrino mass is given by $|m'_\nu| \approx m^2/M_R$. This confirms the standard seesaw suppression mechanism. For instance, if we assume $m \sim 100\,\mathrm{GeV}$ and $|m'_\nu| \sim 0.1\,\mathrm{eV}$, the mechanism necessitates a heavy scale of $M_R \sim 10^{14}\,\mathrm{GeV}$.

4.5.2 The Chiral Phase and CP Violation

A novel feature of this model is that the phase of the light neutrino mass in (28) is directly determined by the chiral angle α

$$\mathrm{Phase}\,(m'_\nu) = e^{-2i\alpha}. \tag{29}$$

This dependence has the following implications:

- **Leptonic CP Violation:** In a three-generation scenario, this complex phase translates into Majorana phases in the PMNS mixing matrix. A value of $\alpha \neq k\pi/2$ implies a novel source of leptonic CP violation.
- **BSM Link:** This establishes a direct link between the theoretical angle α of the CDE and a physical observable mediated by the seesaw scale M_R.

5 Generalization to Higher Spins

To systematically address the extension to higher spins, we adopt the Bargmann-Wigner (BW) formalism, which provides a rigorous group-theoretical framework for constructing relativistic wave equations for particles of arbitrary spin s. Established by V. Bargmann and E. P. Wigner [8], this method posits that the wavefunction of a particle with spin s can be represented by a multispinor field $\Psi_{\alpha_1\alpha_2\ldots\alpha_{2s}}$ of rank $2s$, which effectively describes a composite state of $2s$ fundamental spin-1/2 fermions. To ensure that this object transforms irreducibly under the Poincaré group, the multispinor must satisfy two conditions: it must be totally symmetric under the exchange of any pair of spinor indices, and it must satisfy the Dirac equation for each index independently

$$(i\,\gamma^\mu\partial_\mu - m)_{\alpha_k\beta}\;\Psi_{\alpha_1\ldots\beta\ldots\alpha_{2s}}(x) = 0, \qquad \text{for } k = 1,\ldots,2s. \tag{30}$$

To address the specific extension to vector bosons, we adopt the methodology established by Watson in their generalization of the Proca equation [6]. Their approach serves as the guiding framework for this analysis. Specifically, for the vector boson case ($s = 1$), the formalism begins with the Bargmann-Wigner equations for a symmetric rank-2 spinor $\Psi_{\alpha\beta}$. Crucially, the standard scalar mass m is replaced by the chiral mass operator $\hat{M}$ derived from the fermionic theory, applied to each spinor index independently

$$\left(i\gamma^\mu\partial_\mu - \hat{M}\right)_{\alpha\lambda}\Psi_{\lambda\beta}(x) = 0, \quad \text{with} \quad \hat{M} = me^{-i\alpha\gamma^5}. \tag{31}$$

The wavefunction is then expanded in terms of the Clifford algebra generators. The symmetry constraint intrinsic to the Bargmann-Wigner formalism for $s = 1$ requires the spinor $\Psi_{\alpha\beta}$ to be symmetric, which effectively eliminates the scalar, pseudoscalar, and axial-vector components. Consequently, the wavefunction reduces naturally to a superposition of only the vector (A_μ) and tensor ($F_{\mu\nu}$) sectors

$$\Psi(x) = \left(\gamma^\mu A_\mu(x) + \frac{1}{2}\sigma^{\mu\nu}F_{\mu\nu}(x)\right)C, \tag{32}$$

where C is the charge conjugation matrix. Substituting this ansatz back into the fundamental chiral equations leads to a decoupling of the components. While the standard theory yields the Proca equation $\partial_\mu F^{\mu\nu} + m^2 A^\nu = 0$, the chiral extension results in the Generalized Proca Equation

$$\partial_\mu F^{\mu\nu} + m^2\cos^2(\alpha)\,A^\nu + \cdots = 0. \tag{33}$$

It is important to clarify that the appearance of the $\cos^2(\alpha)$ factor in this sector does not violate the relativistic structure of the theory; rather, it acts as a scaling parameter that defines an effective physical mass $M_{\text{eff}} = m\cos(\alpha)$. This implies that the chiral angle can modulate the observable mass of the boson without altering the underlying geometric consistency of the Poincaré group.

5.1 The Chiral Rarita-Schwinger Equation

Building on this success, we apply the procedure to the Rarita-Schwinger field ($s = 3/2$). To maintain full consistency with the underlying Bargmann-Wigner theory, we treat the field strictly as a totally symmetric rank-3 spinor $\Psi_{\alpha\beta\gamma}$. In this formalism, the field is a composite state of three fundamental spin-1/2 fermions. The chiral mass generation mechanism is applied to each of the three constituent spinor indices. Since the chiral rotation $e^{-i\alpha\gamma^5}$ induces a projection factor of $\cos(\alpha)$ for a single spinor (as seen in the Dirac case), the composite mass term for a rank-3 object must scale as the product of the projections of its constituents. This leads to a cubic scaling law

$$\hat{M}_{\text{composite}} \to m \cos^3(\alpha)\,. \tag{34}$$

Consequently, the modified Chiral Rarita-Schwinger equation takes the form

$$\epsilon^{\mu\nu\rho\sigma}\gamma^5\gamma_\nu\partial_\rho\psi_\sigma + im\cos^3(\alpha)\,\sigma^{\mu\nu}\psi_\nu = 0. \tag{35}$$

This result suggests a general scaling law for the effective mass of a particle with spin s: $M_{\text{eff}}^{(s)} \approx m\cos^{2s}(\alpha)$. This theoretical prediction aligns with the linear scaling found for Dirac fermions ($2s = 1 \to \cos^1$) and the quadratic scaling found for Proca bosons ($2s = 2 \to \cos^2$), offering a unified geometric interpretation of mass modulation across the spin spectrum. Finally, we note that for higher-spin fields, the introduction of non-trivial mass matrices requires careful checking of the constraint algebra ($\gamma^\mu\psi_\mu = 0$ and $\partial^\mu\psi_\mu = 0$) to avoid the propagation of acausal modes. This is a known issue in massive Rarita-Schwinger theories referred to as the Velo-Zwanziger problem [9]. While a full causality analysis is beyond the scope of this proposal, future work must verify the preservation of these constraints under chiral mass modifications.

6 Discussion and Conclusions

The investigation into the Chiral Dirac Equation (CDE) establishes a consistent algebraic framework for introducing chiral degrees of freedom into relativistic wave equations. Our analysis confirms that the introduction of the chiral angle α preserves the fundamental structure of the quantum theory. Explicit derivation demonstrates that the Hamiltonian $H = \gamma^{0k}p_k + \gamma^0 m e^{-i\alpha\gamma^5}$ remains strictly Hermitian, ensuring real energy eigenvalues and unitary time evolution. Consequently, the CDE constitutes a viable extension of the standard Dirac formalism, physically admissible for the description of fermionic fields.

6.1 Chiral Symmetry and Neutrino Phenomenology

The application of the CDE to neutrino physics does not resolves the vanishing mass problem inherent in pure left-handed theories. By incorporating a Majorana mass term, we derived a Chiral Seesaw Mechanism where the effective Dirac mass is modulated by the chiral phase, $M_D = me^{-i\alpha}$. This leads to two distinct physical consequences. First, the active neutrino mass is naturally suppressed by the heavy sterile scale M_R according to $m'_\nu \approx m^2 M_R^{-1} e^{-2i\alpha}$. Second, and perhaps more significantly, the chiral angle α manifests as a macroscopic observable: it determines the phase of the light neutrino mass eigenstate. If $\alpha \neq k\pi/2$, this mechanism provides a geometric origin for leptonic CP violation within the PMNS matrix, linking the internal symmetries of the Poincaré group representation directly to experimental observables beyond the Standard Model.

6.2 Geometric Scaling of Higher-Spin Masses

The extension of this formalism to bosonic sectors suggests a unified geometric scaling law for mass generation. For vector bosons ($s = 1$), the enforcement of chiral symmetry on the constituent spinors of the Bargmann-Wigner multispinor results in an effective mass $M_{\text{eff}} \approx m \cos^2(\alpha)$. Our preliminary analysis of the Rarita-Schwinger field ($s = 3/2$) indicates a continuation of this trend, yielding a cubic scaling $M_{\text{eff}} \approx m \cos^3(\alpha)$. This supports the hypothesis that the effective mass of a particle with spin s scales as $M_{\text{eff}}^{(s)} \approx m \cos^{2s}(\alpha)$, identifying mass modulation as a function of the composite spin structure. However, as noted, the consistency of these higher-spin equations requires rigorous verification against the Velo-Zwanziger constraints to preclude acausal propagation modes.

6.3 Algebraic Generalization via Idempotents

The construction of wave equations for arbitrary spin s via component-based tensor calculus becomes intractable for $s > 1$. A more robust approach relies on the method of Orthogonal Idempotents proposed by Watson and Musielak. Rather than constructing Lagrangians term-by-term, this formalism utilizes generalized projection operators $\hat{P}$ to define physical subspaces. For a composite system of N spinors, the operator is the tensor product of individual helicity projectors

$$\hat{P}_{s_1 \ldots s_N}(\hat{n}) = \bigotimes_{i=1}^{N} \hat{P}_{s_i}(\hat{n}) \,. \tag{36}$$

This method is algebraically superior as it handles chiral constraints intrinsically. By parametrizing the chiral basis of each projector, the effective mass M_{eff} emerges not as an ad-hoc parameter, but as an eigenvalue of the operator acting on the vacuum, $\hat{P}_{\text{total}}\Psi = M_{\text{eff}}\Psi$. This confirms that chiral degrees of freedom are embedded in the representation structure of the Poincaré group. Future work will utilize this idempotent formalism to classify the causal structure of the derived higher-spin fields.

Declarations

Use of Artificial Intelligence

The authors acknowledge the use of Google Gemini as a tool for grammatical refinement, spelling correction, and the detection of syntactical errors during the preparation of this manuscript. All scientific content, mathematical derivations, and physical interpretations were generated exclusively by the authors. The final text has been rigorously reviewed and verified by the authors to ensure accuracy and integrity.

Conflict of Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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